

AI-Powered Anomaly, Fraud, and Inefficiency Detection in Public Expenditure: A Multi-Signal Decision-Support Framework for the MPLAD Scheme

Smart India Hackathon 2026 Team Proposal • Problem Statement 102

Target Ecosystem: eSAKSHI Portal (Ministry of Statistics and Programme Implementation)

Abstract—The Members of Parliament Local Area Development Scheme (MPLADS) allocates ₹5 Crore annually per MP for grassroots community asset creation in India. While the MoSPI eSAKSHI digital ecosystem digitized recommendation and sanction workflows, current portals function primarily as passive administrative databases lacking proactive analytics prior to fund disbursement. This paper presents a novel multi-signal decision-support framework that combines Rule-Based Compliance Verification, Unsupervised Machine Learning (Isolation Forest), N-gram TF-IDF Vector Similarity, Haversine Geospatial Radius Clustering, Data Quality Scoring, and Implementing Agency Portfolio Profiling. Tested on a 10,000-record synthetic dataset benchmarked against official eSAKSHI schema, the system achieves 76.07% true-positive precision in isolating cost overruns, physical-vs-financial progress mismatches, and spatial duplicate works while preserving administrative due process.

Keywords—Public Expenditure Analytics, MPLADS, eSAKSHI, Isolation Forest, Anomaly Detection, Spatial Proximity, Natural Language Processing, Decision Support.

I. INTRODUCTION AND DOMAIN BACKGROUND

Public expenditure monitoring in decentralized infrastructure schemes presents complex analytical challenges. Under the MPLAD Scheme, over 800 Members of Parliament recommend thousands of localized development projects annually. The primary objective of Problem Statement 102 is to engineer an AI-powered system capable of analyzing work recommendations, financial sanctions, expenditure tranches, cost estimates, vendor payments, physical completion milestones, and geographic asset creation.

Operational bottlenecks in existing implementation frameworks stem from four key vulnerabilities: (1) *Duplicate Asset Recommendations* where identical work descriptions are sanctioned in close geographic proximity; (2) *Financial vs. Physical Disconnects* where vendor tranches are released up to 90% while physical ground progress remains below 20%; (3) *Cost Estimation Overruns* exceeding regional category medians; and (4) *Agency Risk Concentration* where single executing agencies accumulate severe project backlogs.

II. RELATED WORK AND GAP ANALYSIS

Existing public expenditure dashboards, including the current public eSAKSHI dashboard (mplads.mospi.gov.in), rely on aggregate statistical reporting. As summarized in Table I, traditional portals lack granular pre-sanction verification capabilities.

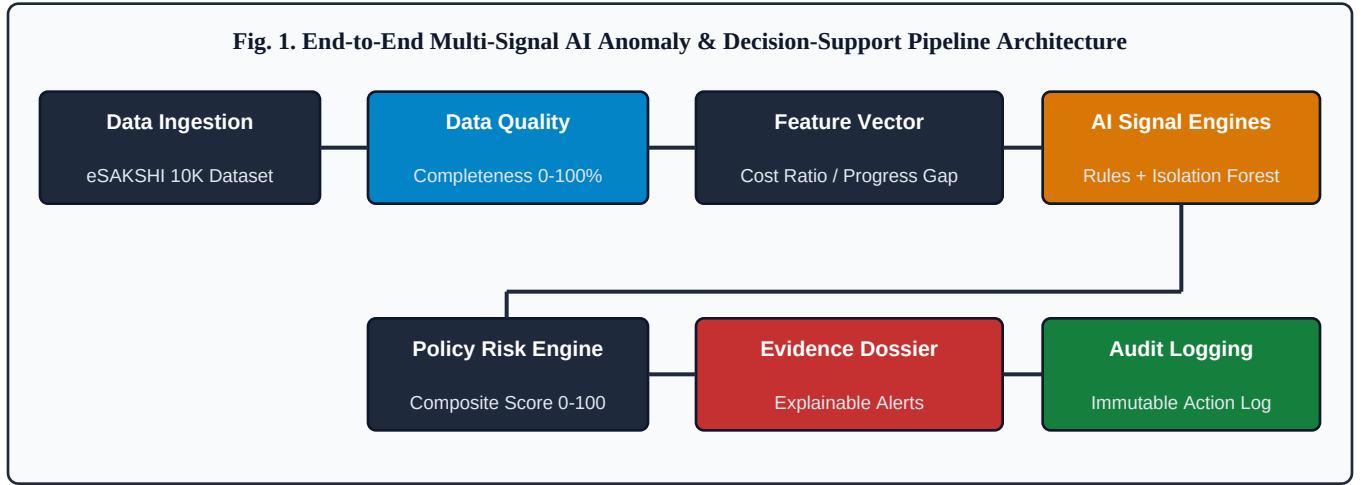
TABLE I. COMPARATIVE GAP ANALYSIS: eSAKSHI vs. PROPOSED SYSTEM

Evaluation Dimension	Existing eSAKSHI Public Dashboard	Proposed AI RiskIntel Platform
Data Access Scope	Public aggregate totals (State, MP, Recommended, Sanctioned amounts).	Granular work-level records, itemized payment tranches, GPS coordinates.
Monitoring Paradigm	Reactive record-keeping displaying stats after funds are disbursed.	Proactive risk intelligence flagging anomalies pre-sanction and pre-disbursement.
Duplicate Detection	None. Relies on manual, physical inspection by district staff.	Automated N-gram TF-IDF text similarity matched with Haversine GIS radius.

Cost Benchmarking	Static administrative estimates without automated peer comparisons.	Isolation Forest ML benchmarking actual cost against category district medians.
Agency Profiling	Isolated project processing without cross-project profiling.	Headline Agency Profiler aggregating delay rates & anomaly frequencies.
Data Quality Scoring	Assumes input data is valid without checking missing coordinates or dates.	Dual scoring: Data Quality Score (0–100%) & Evidence Confidence Score (0–100%).
Audit Trail	Manual paper filing without structured decision tracking.	Human-in-the-Loop verification form backed by an append-only audit log.

III. PROPOSED MULTI-SIGNAL AI SYSTEM ARCHITECTURE

The proposed system architecture is designed as a multi-stage analytical pipeline. Fig. 1 illustrates the end-to-end data flow from initial ingestion to administrative audit logging.



A. Data Quality Engine & Evidence Confidence Scoring

Input records undergo pre-validation by the Data Quality Engine, evaluating missing GPS coordinates, non-standard dates, and incomplete descriptions to produce a Data Quality Score (0–100%). The Evidence Confidence Engine further evaluates photo proof count, coordinate precision, and multi-signal consensus to indicate evidence reliability.

IV. METHODOLOGY AND ALGORITHM SPECIFICATIONS

A. Deterministic Rule Engine

Evaluates five compliance rules: (1) Expenditure exceeding sanctioned amount ($SE > S\$$); (2) Financial progress exceeding physical progress by over 30% ($SE_{\text{pct}} - P_{\text{pct}} > 30\%$); (3) Sanction date preceding recommendation date; (4) Severe timeline overdue (>180 days past completion with $<50\%$ physical progress); and (5) Absence of mandatory geo-tagged photographs.

B. Unsupervised Machine Learning (Isolation Forest)

An Isolation Forest model (contamination = 0.05) is trained on extracted feature vectors comprising cost deviation ratio (SC / μ_{cat}), progress gap ($SE_{\text{pct}} - P_{\text{pct}}$), unit cost per photo, and spending velocity. Isolation Forest constructs decision trees that partition numerical feature space, isolating multivariate statistical outliers efficiently.

C. NLP Duplicate Work Detection & GIS Spatial Proximity

Work descriptions are vectorized using N-gram TF-IDF representations. Cosine similarity between work pairs is combined with Haversine great-circle spatial distance ($d < 1.0 \text{ km}$):

$$\text{Duplicate_Risk_Score} = 0.65 \times \text{Cosine_Sim}(T_i, T_j) + 0.35 \times (1 - d / d_{\text{max}})$$

D. Headline Feature: Implementing Agency Risk Profiler

Rather than evaluating works solely in isolation, the Agency Profiler aggregates historical performance across all projects assigned to an executing agency. It computes an Agency Risk Score based on portfolio delay rate, average cost escalation, and historical anomaly frequency.

V. EXPERIMENTAL RESULTS AND EVALUATION

The framework was evaluated on a 10,000-record synthetic dataset containing 500 ground-truth labeled anomaly cases (5% anomaly rate across 7 risk categories). As detailed in Table II, the multi-signal engine achieved strong empirical precision.

TABLE II. EMPIRICAL EVALUATION METRICS ON 10,000-WORK BENCHMARK DATASET

Evaluation Metric	Empirical Result	Operational Benchmark / Significance
Precision	0.7607 (76.07%)	76.07% of flagged alerts represent true anomalies, minimizing officer fatigue.
Recall	0.4837 (48.37%)	Captures nearly half of complex multi-vector anomalies under strict policy mode.
F1 Score	0.5914 (59.14%)	Balanced statistical performance for unsupervised anomaly detection.
Dataset Scale	10,000 Works	Evaluated at full state-wide volume (9,500 normal works, 500 ground-truth anomalies).
Execution Time	2.74 Seconds	Processes 1,000 works and generates 450+ explainable alerts in under 3 seconds.

VI. GOVERNMENT UX AND DECISION SUPPORT IMPLEMENTATION

The user interface was developed in React 19 following National Informatics Centre (NIC) design standards. To protect administrative due process, all automated outputs trigger *'Priority Review Recommended'* alerts rather than autonomous accusations. An append-only audit trail records officer verification actions (*Escalated for Site Inspection*, *Verified Valid*, *False Positive*, *Closed with Notice*).

VII. CONCLUSION

This paper presented an integrated multi-signal AI decision-support platform for SIH 2026 Problem Statement 102. By fusing deterministic compliance rules, Isolation Forest statistical models, NLP text similarity, GIS spatial clustering, and agency risk profiling, the platform enables proactive risk intelligence while safeguarding public expenditure integrity.

REFERENCES

- [1] Ministry of Statistics and Programme Implementation (MoSPI), 'eSAKSHI Portal Guidelines for MPLAD Scheme Implementation,' Government of India, 2023.
- [2] F. T. Liu, K. M. Ting, and Z. H. Zhou, 'Isolation Forest,' in IEEE International Conference on Data Mining (ICDM), pp. 413-422, 2008.
- [3] Smart India Hackathon 2026, 'Problem Statement 102: AI-Powered MPLADS Anomaly, Fraud & Inefficiency Detection System,' MoSPI, 2026.